AI Agents and Tool Use

1. What is an AI Agent?

An AI agent is an application in which a language model can determine actions to take using a set of available tools or capabilities. The model interprets a request, selects an appropriate action, receives the result, and uses that result to produce a response.

2. LLM and Agent

An LLM can generate language by itself, while an agent system gives the model access to tools and a control loop. The LLM can therefore act as the decision-making language component, while tools provide access to external information or operations.

3. Tools

A tool can expose a specific operation to the agent. Examples include a knowledge base search function, calculator, database query, weather service, or application API. A well-defined tool describes what it does and what inputs it expects.

4. Knowledge Base Tool

A RAG application can expose its retrieval pipeline as a tool such as knowledge_base_search. The agent can call the tool when a question requires information from the application's documents. The tool returns relevant chunks, which are then provided to the LLM for answer generation.

5. Agent Workflow

A simple agent workflow is: receive the user request, interpret the request, decide whether a tool is needed, call the selected tool, inspect the returned information, and generate a final response. More advanced systems can use multiple tools or multiple reasoning and action steps.

6. Grounding and Safety

Agentic systems should constrain tools and instructions so that the model has clear boundaries. For a knowledge-base assistant, the application can require answers to be grounded in retrieved documents and can require the agent to acknowledge when the available evidence is insufficient.